

# Prototype Of Operational Security and Sharia Compliance Risk for Ethereum Staking

1<sup>st</sup> Rinda Hesti Kusumaningtyas  
Department of Information Systems  
State Islamic University Syarif  
Hidayatullah Jakarta  
Jakarta, Indonesia  
rinda.hesti@uinjkt.ac.id

2<sup>nd</sup> Fitroh Fitroh  
Department of Information Systems  
State Islamic University Syarif  
Hidayatullah Jakarta  
Jakarta, Indonesia  
fitroh@uinjkt.ac.id

3<sup>rd</sup> Maulana Asykari Muhammad  
Department of Information Systems  
State Islamic University Syarif  
Hidayatullah Jakarta  
Jakarta, Indonesia  
maulana.asykari22@mhs.uinjkt.ac.id

4<sup>th</sup> Nur Aeni Hidayah  
Department of Information Systems  
State Islamic University Syarif  
Hidayatullah Jakarta  
Jakarta, Indonesia  
nur.aeni@uinjkt.ac.id

5<sup>th</sup> Sarip Hidayatuloh  
Department of Information Systems  
State Islamic University Syarif  
Hidayatullah Jakarta  
Jakarta, Indonesia  
sarip\_hidayatuloh@uinjkt.ac.id

**Abstract**—Ethereum’s transition to Proof-of-Stake has expanded opportunities for users to participate in digital asset staking. However, different staking architectures expose users to distinct operational, technical, financial, and Sharia-related risks. This study develops a qualitative document-based comparative framework to assess Solo Staking and decentralized Liquid Staking through Lido Finance and to translate the findings into an interface-based risk disclosure design. The study was conducted in four phases: problem identification and criteria definition, evidence collection and architecture analysis, multidimensional risk assessment, and interface-based mitigation design. The analysis applied PIECES framework and Fishbone analysis to categorize the issues in the identified problems and to analyze the possible problems related to *gharar*, *maysir*, *riba*, *dharar*, and *hifz al-mal* using the parameters of *Fiqh Muamalah*. We then modeled the proposed user interactions and disclosure mechanisms using UML use case diagrams and interface prototypes. The results show that Solo Staking retains a one-to-one correspondence between validator duties and protocol rewards, but shifts the responsibility of node availability, key management, infrastructure upkeep and withdrawal management to individual validators. Lido Finance lowers the minimum capital and technical participation requirements and offers tokenized liquidity. But it introduces additional dependencies on smart contracts, validator operators, token accounting, secondary-market pricing, and DeFi composability. The Sharia implications of liquid staking depend on the contract structure and how it is used, especially when liquid staking tokens are used for lending, leverage, recursive-yield or speculative activities. From these findings, the proposed interface includes a contractual acknowledgment based on *Wakalah bil Istithmar*, information transparency about the transaction, indicators of risk, and a dashboard for monitoring risks called the Risk Explorer. This research postulates a comparative risk taxonomy and a conceptual interface framework to enhance disclosure, informed consent, operational risk awareness and Sharia-oriented governance in Ethereum staking.

**Keywords**— *Ethereum Staking, operational risk, smart contract security, Liquid Staking, Islamic fintech.*

## I. INTRODUCTION

Global participation in crypto-assets continues to expand. The Crypto.com Research and Insights Team estimated that the number of global cryptocurrency owners increased from

583 million in January 2024 to 659 million in December 2024 [1]. Of this total, approximately 142 million people owned Ether, representing 21.7% of global crypto owners. Crypto ownership is also concentrated among younger users, indicating that Generation Z and Millennials form a substantial segment of the digital-asset market. In Indonesia, crypto-assets are legally recognized as tradable digital financial assets but cannot be used as legal tender [2], [3]. At the same time, the Ijtima Ulama of the Indonesian Ulema Council stated that cryptocurrency as currency is prohibited, while its trading as a digital commodity requires compliance with *sil’ah* criteria and the absence of *gharar*, *dharar*, and *qimar* [4].

This regulatory and jurisprudential distinction creates uncertainty for Muslim investors who participate in digital-asset markets. The Majelis Tarjih and Tajdid of Muhammadiyah has classified utility-based crypto-assets as *mal muta’awwam*, provided that the asset has identifiable utility and its transactions do not involve market manipulation, interest-bearing mechanisms, or prohibited speculation [6]. However, this conditional recognition does not automatically determine the Sharia status of staking activities. Ethereum staking involves validator duties, reward mechanisms, slashing penalties, liquidity constraints, and smart contract dependencies that require separate assessment. From a Sharia perspective, the management of these risks is related to *hifz al-mal*, particularly where technical failures or unclear loss allocation may expose users to avoidable financial harm [6]. Habib and Jawa explain that validator participation in Proof-of-Stake systems may be examined through *ju’alah* because rewards are linked to identifiable validation work [7].

Although these ecosystem components have been extensively assessed in the existing literature, they have been done so in isolation. Research from Adamyk et al. developed a DeFi risk framework covering financial, operational, technological, and compliance-related vulnerabilities [8]. Habib and Jawa assessed Proof-of-Stake validation through the concept of *ju’alah* [7]. Izadin et al. linked digital-asset risk mitigation with *Maqasid al-Shariah* and wealth protection [6]. Xiong et al. identified liquidity, leverage, and cascading liquidation risks associated with liquid staking derivatives, including the use of stETH in external DeFi protocols [9]. However, these studies do not integrate Sharia

screening, validator risk, smart contract governance, liquidity exposure, and user-facing disclosure into a single comparative assessment of Solo Staking and Lido Finance.

Therefore, this study conducts a qualitative document-based comparison of Solo Staking and Liquid Staking through Lido Finance using four phases: criteria definition, evidence and interface observation, multidimensional assessment, and interface-based mitigation recommendation. The results show that Solo Staking provides a direct relationship between validation duties and rewards but transfers infrastructure and key-management responsibilities to the validator. Lido Finance lowers participation barriers and provides tokenized liquidity but adds smart contract, operator, market, and composability dependencies. The main contribution is a taxonomy that separates Sharia screening from operational, technical, and financial scoring, applies it comparatively to two staking models, and translates the findings into interface recommendations.

## II. RELATED WORKS

### A. Ethereum Staking Architectures

The transition of the Ethereum network to a Proof-of-Stake (PoS) consensus mechanism has significantly transformed its economic and security framework. According to Brummer [10], cryptocurrency staking models vary based on capital commitment, technical execution, and information asymmetry, dividing the ecosystem into independent and delegated validation. Solo staking represents the foundation of this system, where participants independently deploy physical node infrastructures and deposit a strict threshold of 32 ETH to perform core consensus duties, including block proposal, attestation, and participation in network finality.

Conversely, delegated staking models reduce these high capital and technical barriers via cryptographic pooling. Research from Ries points out that the deployment of Liquid Staking Derivatives (LSDs) enables users to aggregate smaller amounts of capital while receiving liquid token representations that garner rewards [11]. Furthermore, centralized entities have integrated these delegation mechanisms through custodial staking. As described by Röckener, custodial frameworks allow users to delegate asset custody entirely to centralized exchanges (CEXs) in exchange for automated rewards, creating a persistent structural tension between decentralized network principles and operational centralization [12].

### B. Sharia Compliance Parameters

A prior systematic literature review of 32 peer-reviewed studies identified recurring Sharia parameters relevant to crypto staking and decentralized finance, including *gharar*, *maysir*, *riba*, *dharar*, *hifz al-mal*, *ju'alah*, and *ujrah* [13]. The review indicates that Sharia compliance cannot be determined solely from the existence of financial returns. The assessment must also consider the contractual basis of the reward, the clarity of rights and obligations, the allocation of losses, the source of yield, and the extent of speculative exposure [7], [13]. Based on this synthesis, the present study operationalizes these principles as comparative

criteria for Solo Staking and Liquid Staking through Lido Finance.

TABLE I. SHARIA ANALYTICAL BASIS AND TECHNICAL INTERPRETATION

| Principle                    | Descriptions                                               | Technical assessed                                                                      | Literatures     |
|------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------|
| <b>Maysir</b>                | Prohibition of gambling and dominant speculation           | LST trading, leverage, looping, secondary-market exposure                               | [9], [14]       |
| <b>Riba</b>                  | Prohibition of stipulated increase in debt-based exchange  | Lending integrations, guaranteed yield, debt-like return structures                     | [7], [9], [15]  |
| <b>Gharar</b>                | Prohibition of material contractual uncertainty            | Reward source, fees, slashing allocation, withdrawal conditions, smart-contract opacity | [16], [17]      |
| <b>Dharar</b>                | Prevention of unjustified harm                             | Slashing, hacks, key compromise, downtime, concentration                                | [6], [18], [19] |
| <b>Hifz al-Mal</b>           | Protection and preservation of wealth                      | Monitoring, risk disclosure, alerts, exit information                                   | [6], [13], [16] |
| <b>Ju'alah</b>               | Reward for completing a defined task or achieving a result | Validator attestation and block proposal                                                | [7]             |
| <b>Wakalah bil Istithmar</b> | Delegated investment management under specified authority  | Proposed liquid-staking e-contract                                                      | [20]            |

## III. METHODOLOGY

### A. Research Design and Data Acquisition

This study employs a qualitative document-based comparative assessment. The analysis is limited to Solo Staking and decentralized liquid staking through Lido Finance. Custodial staking is excluded because it introduces centralized custody and intermediary risks beyond the scope of this study. Evidence was collected from Ethereum and Lido documentation, smart contract audits, validator analytics, protocol dashboards, interface observations, and peer-reviewed Sharia-finance literature.

### B. Four-Phase Comparative Evaluation Framework

Existing studies provide complementary but fragmented foundations for evaluating Ethereum staking architectures. Adamyk et al. classify decentralized finance risks into operational, technological, financial, and compliance-related dimensions [8]. Muhammad et al. identify Sharia parameters relevant to crypto staking, including *gharar*, *maysir*, *riba*, *dharar*, and *hifz al-mal* [13]. Brummer further emphasizes the importance of disclosure in reducing information asymmetry within blockchain-based systems [10]. These studies provide the conceptual basis for integrating risk identification, Sharia screening, comparative assessment, and user-facing disclosure into a single analytical procedure. Based on these foundations, this study synthesizes a four-phase comparative evaluation framework Fig. 1 illustrate those framework.

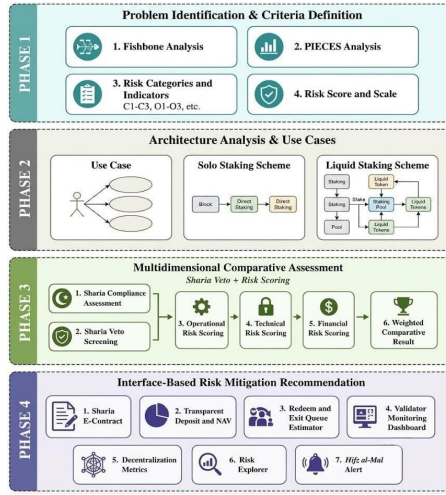


Fig. 1. Four-phase comparative evaluation framework.

## IV. RESULT

### A. Problem Identification & Criteria Definition

Phase 1 defines the comparison objects, assessment dimensions, indicators, and decision rules used to evaluate Solo Staking and Liquid Staking through Lido Finance. Habib and Jawa explain that staking models differ in terms of validator participation, reward mechanisms, asset control, and contractual relationships [7]. Adamyk et al. further classify DeFi risks into operational, technological, financial, and compliance-related dimensions. Based on these considerations, this study evaluates both staking models across Sharia compliance, operational risk, technical risk, and financial risk [8].

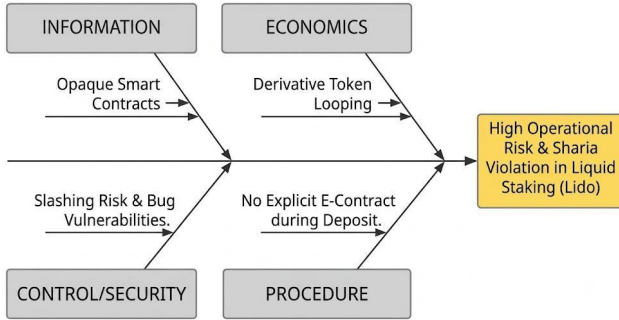


Fig. 2. Fishbone Analysis on Liquid Staking

Figure 2 identifies the main sources of operational and Sharia-related risk in liquid staking, covering information, economic, control/security, and procedural factors. These factors are further classified through the PIECES framework in Table II.

TABLE II. PIECES FRAMEWORK ANALYSIS ON CONVENTIONAL LIQUID STAKING

| Category    | Identified Problem                                                           | Impact                                                             |
|-------------|------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Performance | Validator node downtime and network latency are not transparently mitigated. | Triggers algorithmic slashing penalties, risking investor capital. |

|             |                                                                                       |                                                                                 |
|-------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Information | Black-box smart contract logic and hidden validator metrics during deposit.           | Causes information asymmetry, triggering Algorithmic <i>Gharar</i> .            |
| Economics   | Issuance of liquid staking tokens (stETH) that are used in recursive DeFi strategies. | Decouples profit from real utility, inducing <i>Maysir</i> .                    |
| Control     | Susceptibility to reentrancy bugs, code exploits, and oracle manipulation.            | Single points of failure may affect asset protection under <i>hifz al-mal</i> . |
| Efficiency  | Absence of integrated Sharia filtering forces users to manually verify contracts.     | Increases the information burden and risk exposure for retail users.            |
| Service     | No explicit digital E-Contracts (Wakalah) provided during the onboarding process.     | May limit the clarity of mutual consent in the transaction.                     |

The Sharia dimension is assessed separately from the numerical operational, technical, and financial dimensions. *Maysir*, *riba*, and *gharar* function as screening parameters, while *dharar* and *hifz al-mal* provide supporting principles for harm prevention and asset protection [6], [13], [14]. Operational, technical, and financial indicators are scored to describe the relative exposure of each staking model. This separation is intended to prevent favorable technical performance from offsetting a material Sharia concern. As elucidated by Adamyk et al and Ajayi et al., these categories correspond to risks commonly identified in blockchain and DeFi environments, including smart contract defects, key compromise, infrastructure failure, market exposure, and liquidity constraints [8], [18]. Compliance risk evaluates whether the assessed configuration contains or facilitates elements associated with *maysir*, *riba*, or *gharar*. The taxonomy functions as an analytical classification and not as a final Sharia ruling.

The relationship among the dimensions is not necessarily one-to-one. Yaqin et al., explain that blockchain transparency may reduce some forms of information asymmetry while creating new technical and interpretive uncertainties [16]. Research from Brummer similarly notes that blockchain information may be publicly available but remain difficult for users to understand [10]. Therefore, an undisclosed smart contract dependency may increase contractual uncertainty, while a security failure or key compromise may result in financial loss. The taxonomy is used to trace these relationships between technical conditions, financial exposure, and Sharia-related assessment.

TABLE III. RISK CATEGORIES AND INDICATORS

| ID | Indicators                | Data Source                          | Risk Categories   |
|----|---------------------------|--------------------------------------|-------------------|
| O1 | Uptime                    | Rated.Network API                    | Operational Risks |
| O2 | Slashing History          | Beaconcha.in API                     |                   |
| O3 | Client Diversity          | Dune Analytics / ClientDiversity.org |                   |
| T1 | Smart Contract Security   | Etherscan / Audit Reports            | Technical Risks   |
| T2 | Key Management            | Infrastructure Specifications        |                   |
| T3 | Infrastructure Redundancy | Server Specifications                |                   |
| F1 | Liquidity Constraint      | ValidatorQueue API / Beaconcha.in    | Financial Risk    |
| C1 | Maysir Potential          | Smart Contract & Whitepaper          | Compliance Risks  |

|    |                  |                                |  |
|----|------------------|--------------------------------|--|
| C2 | Riba Potential   | Token Architecture & Ecosystem |  |
| C3 | Gharar Potential | Public API Availability        |  |

Table III specifies the indicators used to compare the two staking models. Chen and Golab explain that Ethereum validators must maintain network connectivity and perform protocol duties to earn rewards and avoid penalties [19]. Therefore, validator uptime and slashing history are included as operational indicators. Ajayi et al. identify smart contract vulnerabilities, key compromise, and infrastructure weaknesses as cybersecurity concerns in blockchain-based financial systems [18]. These findings support the inclusion of smart contract security, key management, and infrastructure redundancy as technical indicators.

The compliance indicators do not assume that every staking reward, token, or liquidity mechanism is prohibited. *Maysir* potential is assessed when returns depend mainly on speculative or leveraged exposure rather than identifiable validation services [9], [14]. *Riba* potential is examined when returns arise from lending, debt-like obligations, or stipulated additions, rather than from compensation for validation work [7], [17]. *Gharar* potential is assessed through the clarity of contractual rights, reward sources, fees, penalties, token mechanics, and withdrawal conditions [10], [16].



Fig. 3. Risk Measurements Criteria

Figure 3 presents the measurement rules used in the assessment. Muhammad et al. distinguish Sharia-related parameters from conventional technological and financial risk indicators [13]. Accordingly, Sharia indicators are not included in the aggregate operational, technical, and financial score because they serve as categorical screening criteria. A material Sharia concern is recorded as a review or veto condition, while the remaining indicators are scored comparatively to indicate the relative exposure of the two staking models. The score interpretation is presented in Table IV.

TABLE IV. RISK SCORE AND SCALE

| Score          | Scale             |
|----------------|-------------------|
| 7 – 11         | Acceptable Risk   |
| 12 – 16        | Tolerable Risk    |
| 17 – 21        | Unacceptable Risk |
| Special Clause | Sharia Veto Right |

### B. Architecture Analysis & Use Cases

Phase 2 collects documentary, technical, operational, financial, Sharia-related, and interface-based evidence for the comparison. Technical evidence was obtained from Ethereum and Lido documentation, smart contract repositories, audit reports, validator analytics, and blockchain monitoring services. Operational evidence includes validator uptime, slashing history, client or operator concentration, key-management architecture, and infrastructure redundancy. Financial evidence covers withdrawal conditions, exit-queue exposure, liquidity access, token conversion, rebasing, and secondary-market use. The selection of these sources follows prior DeFi research that combines documentation, on-chain data, protocol analytics, and monitoring platforms to evaluate technical and financial risk [9], [11]. Sharia-related evidence was derived from *Fiqh Muamalah* literature, *Maqasid al-Shariah* principles, crypto-asset fatwas, and the authors' prior systematic literature review [7], [13], [15].

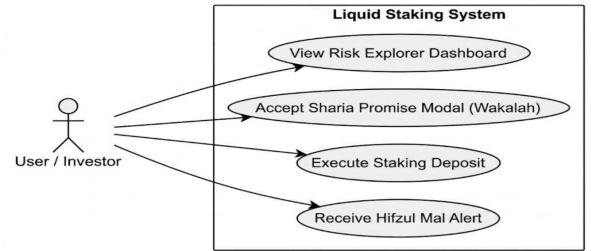


Fig. 4. Use case in Liquid Staking

Figure 4 presents the main user interactions in the proposed liquid staking system. The user can view the Risk Explorer dashboard, review and accept the Wakalah-based agreement, execute a staking deposit, and receive a hifz al-mal alert. These use cases define the functional scope of the proposed interface layer.

Solo Staking is the direct connection of the validator to Ethereum's consensus layer as shown in Figure 3. The validator deposits 32 ETH, holds the validation and withdrawal keys, maintains the node infrastructure and is rewarded/penalized based on the execution of protocol duties. The Ethereum validators have to be online and perform actions such as block proposal and attestation, with penalties or rewards tied to protocol performance [7], [19]. This reduces the dependency on a separate staking provider, but the burden of key management, uptime, software management and infrastructure is passed on to the individual validator. Liquid Staking through Lido Finance adds a protocol layer between the user and underlying validators.

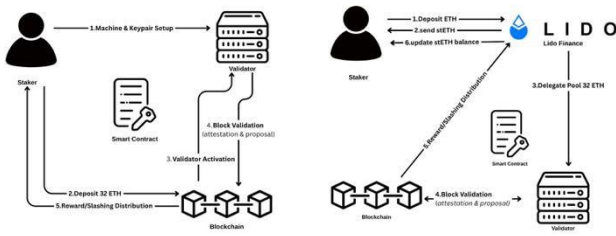


Fig. 5. Solo and Liquid staking interaction scheme.

Figure 5 compares the interaction structures of Solo Staking and Liquid Staking.

### C. Multidimensional Risk Assessment

To better visualize the comparison in Phase 3, this research employs a spider chart to evaluate the multidimensional risk profiles of Liquid Staking via Lido Finance against Solo Staking, as illustrated in Figure 7.

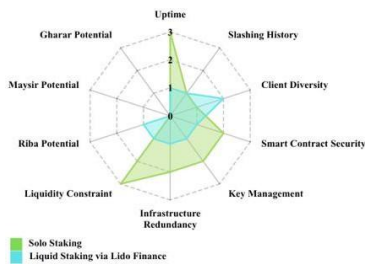


Fig. 6. Comparative risk profile of Lido Finance and Solo Staking.

Figure 6 compares the risk profiles of Lido Finance and Solo Staking. Lido Finance lowers participation requirements through pooled validation, tokenized liquidity, and managed infrastructure, but adds dependencies related to smart contracts, validator operators, token accounting, secondary-market pricing, and DeFi composability [9], [10]. Solo Staking maintains a direct relationship between validation duties and protocol rewards, which may support a conditional analogy with *ju'alah* or *ujrah* [7], [19]. However, the validator remains responsible for node availability, software maintenance, key security, infrastructure redundancy, and withdrawal management.

### D. Interface-Based Risk Mitigation

Based on the comparative assessment, the proposed interface combines Sharia, operational, technical, and financial information within a monitoring environment. The interface presents validator performance, smart contract status, liquidity exposure, and Sharia-related alerts to support disclosure, informed consent, and *hifz al-mal* oriented monitoring.

The proposed interface also provides a location for contractual acknowledgment and risk disclosure. The design does not modify the underlying Ethereum or Lido protocols but functions as an information and decision-support layer. Its purpose is to reduce information asymmetry by presenting technical and financial information in a form that can be used during transaction decisions [8], [10]. The contractual component is presented as a design reference and should not be interpreted as evidence that the current Lido protocol formally adopts a specific Sharia contract. The use of *Wakalah bil Istithmar* in this study is limited to clarifying authorization, asset management, fees, and the distribution of responsibilities in the proposed interface [20].

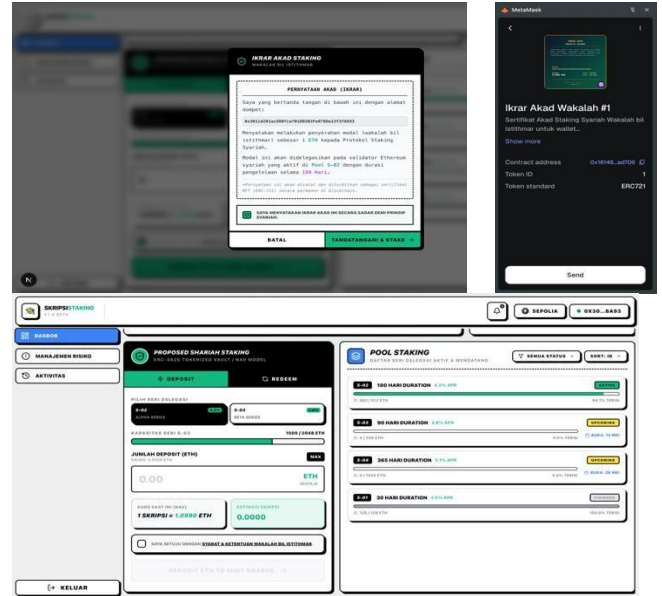


Fig. 7. Proposed transparent deposit interface integrating NAV, fee, conversion, and risk information.

Figure 7 presents the proposed deposit interface, combining contract acknowledgment, net asset value, conversion value, fees, estimated output, and selected risk indicators. The interface functions as an information and decision-support layer without modifying the Ethereum or Lido protocols [8], [10]. *Wakalah bil Istithmar* is used as a design reference to clarify authorization, asset management, fees, and the allocation of responsibilities, rather than as evidence that Lido currently applies a specific Sharia contract [20].

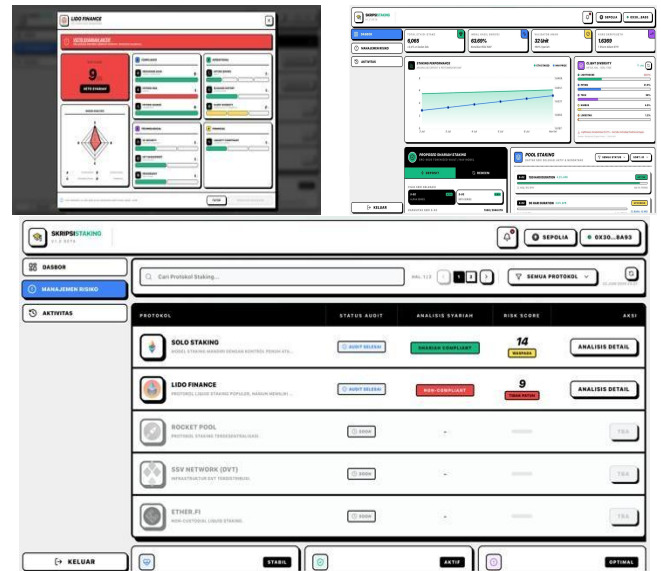


Fig. 8. Proposed dashboard interfaces featuring a validator-ranking system, a performance monitor, and a multidimensional Risk Explorer

Figure 8 presents the supporting dashboard interfaces, consisting of validator ranking, performance monitoring, and multidimensional risk assessment. These components provide information before and after staking transactions but do not replace review by Sharia and technical experts.

## V. CONCLUSION

This research tries to build a comparative taxonomy to assess Solo Staking and Liquid Staking via Lido Finance on operational, technical, financial and Sharia-related aspects. The framework distinguishes between Sharia screening and aggregate risk scoring, as a material contractual prohibition cannot be compensated by less technical exposure. The score is thus understood as a relative comparison and not an industry certified risk rating.

Solo Staking establishes a direct relationship between validation activities and protocol rewards, supporting a conditional analogy with *ju'alah* or *ujrah*. But the model puts node availability, crucial security, infrastructure redundancy, and withdrawal management on the individual validator. Lido Finance lowers minimum capital and technical hurdles, and offers a transferable representation of staked ETH, but introduces dependencies on intelligent contracts, validator operators, token accounting, secondary market pricing, and DeFi composability. These features require configuration-specific Sharia assessment and do not justify a universal ruling on liquid staking.

The proposed interface layer translates the taxonomy into disclosure, monitoring, and early-warning functions. The intent is to provide user-friendly representation of contractual details, validator prerequisites, liquidity exposure and Sharia review indicators. The interfaces are yet conceptual artifacts that need to be validated by Sharia experts, cybersecurity practitioners and end users before they are put into operation.

## ACKNOWLEDGMENT

The authors would like to thank the Department of Information Systems, State Islamic University Syarif Hidayatullah Jakarta, for academic support during the preparation of this study.

## REFERENCES

- [1] Crypto.com, "Crypto Market Sizing Report 2024," 2024. [Online]. Available: <https://crypto.com/en/research/2024-crypto-market-sizing-report>
- [2] A. Febrianto, Ismail, and D. Iryani, "Kepastian Hukum Cryptocurrency Sebagai Alat Pembayaran Di Indonesia," *The Juris*, vol. 9, no. 2, pp. 430–439, 2025, doi: 10.56301/juris.v9i2.1897.
- [3] A. Kindi, "Perlindungan Hukum Pelanggan Aset Kripto Terhadap Investasi Aset Kripto Asixv2V Berdasarkan Peraturan Bappebti Nomor 4 Tahun 2023," Universitas Islam Indonesia, 2023. [Online]. Available: <https://dspace.uui.ac.id/handle/123456789/46572>
- [4] M. Rizieq, "Keabsahan Kripto dalam Perspektif Hukum Ekonomi Syariah: Studi Kasus Fatwa DSN-MUI terhadap Aset Kripto," *Jurnal Multidisiplin Ilmu Akademik*, vol. 2, no. 4, pp. 61–72, 2025, [Online]. Available: <https://doi.org/10.61722/jmia.v2i4.5050>
- [5] F. Majelis, T. Dan, and P. P. Muhammadiyah, "Fatwa Majelis Tarjih Dan Tajdid Pimpinan Pusat Muhammadiyah," Mar. 2026, *Yogyakarta*.
- [6] A. A. I. Izadin, R. Mohd. Yusof, and A. R. Mazlan, "The integration of Maqasid Shariah in evaluating stablecoins and traditional cryptocurrencies for Islamic portfolios diversification," *International Journal of Islamic and Middle Eastern Finance and Management*, vol. 18, no. 3, pp. 577–597, 2025, doi: 10.1108/IMEFM-08-2024-0380.
- [7] F. Habib and A. Jawa, "Crypto staking from the Shariah perspective," in *Islamic Finance in the Digital Age*, Edward Elgar Publishing, 2024, pp. 150–164. doi: 10.4337/9781035322954.00019.
- [8] B. Adamyk, V. Benson, O. Adamyk, and O. Liashenko, "Risk Management in DeFi: Analyses of the Innovative Tools and Platforms for Tracking DeFi Transactions," *Journal of Risk and Financial Management*, vol. 18, no. 1, p. 38, 2025, doi: 10.3390/jrfm18010038.
- [9] X. Xiong, Z. Wang, X. Chen, W. Knottenbelt, and M. Huth, "Leverage Staking with Liquid Staking Derivatives (LSDs): Opportunities and Risks," in *2025 IEEE International Conference on Blockchain and Cryptocurrency, ICBC 2025*, 2025, pp. 1–9. doi: 10.1109/ICBC64466.2025.11114596.
- [10] C. Brummer, "A Developer Theory of Disclosure," *Stanford J. Blockchain Law Policy*, Jan. 2026.
- [11] A. Novocký, K. Košťál, and M. Ries, "SoK: Comprehensive Analysis of Token Allocations, Distributions, and Their Effect on Token Value and User Participation," *Lecture Notes in Operations Research*, vol. Part F1470, pp. 224–262, 2026, doi: 10.1007/978-3-032-13377-9\_11.
- [12] M. Röckener, *MASTER THESIS Centralized Exchanges Incorporating Decentralized Features: Factors Driving Implementation and Impact on the Competitive Landscape*. monami.hs-mittweida.de, 2024. [Online]. Available: [https://monami.hs-mittweida.de/files/15457/BC21w1-M\\_Masterthesis\\_Röckener.pdf](https://monami.hs-mittweida.de/files/15457/BC21w1-M_Masterthesis_Röckener.pdf)
- [13] M. A. Muhammad, Fitroh, and R. H. Kusumaningtyas, "Risk Management and Sharia Compliance in Crypto Staking: A Systematic Literature Review," *Journal of Applied Informatics and Computing (JAIC)*, vol. 10, no. 4, 2026.
- [14] N. M. P. Rahmadani and F. Fajar, "Aligning Cryptocurrencies With Islamic Law: Challenges, Ethical Concerns, and Regulatory Solutions," *Istinbath*, vol. 23, no. 2, pp. 366–383, 2024, doi: 10.20414/ijhi.v23i2.887.
- [15] Majelis Tarjih dan Tajdid PP Muhammadiyah, "Fatwa Majelis Tarjih dan Tajdid Pimpinan Pusat Muhammadiyah tentang Kripto Sebagai Aset Keuangan Digital," 2026, *Yogyakarta*.
- [16] M. A. Yaqin, S. K. Adriani, and N. Kholis, "Algorithmic Trust and Digital Gharar: A Maqāṣidiyyah Framework for Evaluating Blockchain in Islamic Trade Systems," *Int. J. Econ. Manag. Account.*, vol. 3, no. 1, pp. 79–90, 2025, doi: 10.61132/ijema.v3i1.1072.
- [17] Dewan Syariah Nasional MUI, "Fatwa Dewan Syariah Nasional Majelis Ulama Indonesia No. 116/DSN-MUI/IX/2017 tentang Uang Elektronik Syariah," 2017.
- [18] O. O. Ajayi, C. E. Alozie, O. A. Abieba, J. I. Akerele, and A. Collins, "Blockchain Technology and Cybersecurity in Fintech: Opportunities and Vulnerabilities," *Int. J. Sci. Res. Comput. Sci. Eng. Inf. Technol.*, vol. 11, no. 1, pp. 1334–1345, 2025, doi: 10.32628/cseit25111210.
- [19] C. C. Chen and W. Golab, "A game theoretic analysis of validator strategies in Ethereum 2.0," in *Proceedings of the 6th ACM International Symposium on Blockchain and Secure Critical Infrastructure*, 2024, pp. 1–14.
- [20] N. I. Sitepu and S. Maisyura, "Kesesuaian Akad Wakalah bi al-Istitsmar pada Sukuk Tabungan ST009: Ditinjau dari Fatwa DSN-MUI No. 127/DSN-MUI/VII/2019," *Peran Fatwa*, vol. 1, no. 1, p. 227, 2023.